विद्ययाऽमृतमश्नुते

काशी हिन्दू विश्वविद्यालय

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Database Languages and Interfaces

Communicating with Data: Connections, Queries

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Learning Outcomes

- By the end of our session, we will be able to:
- ✱ Understand different database languages.
- ✱ Differentiate between DDL, DML, DQL, DCL, and TCL.
- ✱ Explain various database interfaces.
- ✱ Understand embedded SQL and APIs.
- ✱ Describe web-based database access.
- ✱ Appreciate modern database access through ORMs and REST APIs.

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Why Do We Need Database Languages?

- A DBMS performs many different operations. For example, we may need to:
  - ⁕ Create a database
  - ⁕ Create tables
  - ⁕ Insert records
  - ⁕ Retrieve information
  - ⁕ Modify data
  - ⁕ Delete records
  - ⁕ Control access
  - ⁕ Manage transactions
- <u>Definition</u>
- A Database Language is a specialized language used to define, manipulate, control, and manage data stored in a database.
- *Example*
  - University Database
  - Administrator creates tables.
  - Faculty updates marks.
  - Students retrieve results.
- Each operation uses a different category of database language.

Classification of Database Languages

- Database languages are commonly divided into five categories:
- 1. Data Definition Language (DDL)
- 2. Data Manipulation Language (DML)
- 3. Data Query Language (DQL)
- 4. Data Control Language (DCL)
- 5. Transaction Control Language (TCL)
- SQL
- │
- ├── DDL
- ├── DML
- ├── DQL
- ├── DCL
- └── TCL
- *Note:* In many DBMSs, DQL is considered a part of DML, but separating it helps in learning.

Data Definition Language (DDL)

- <u>Definition</u>
- DDL is used to define and modify the structure of database objects.
- It changes the schema, not just the data.
- *Common Commands*
    - ⁕ CREATE
    - ⁕ ALTER
    - ⁕ DROP
    - ⁕ TRUNCATE
    - ⁕ RENAME
- *Examples*
    - Create a Student table.
    - Add a new column "Email".
    - Delete an unused table.
- *Characteristics*
    - ⁕ Defines database structure.
    - ⁕ Executed by DBAs or designers.
    - ⁕ Affects metadata.

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Data Manipulation Language (DML)

- <u>Definition</u>
- DML is used to insert, update, and delete records stored in database tables.
- *Common Commands*
  - ⁕ INSERT
  - ⁕ UPDATE
  - ⁕ DELETE
- *Example*
  - Insert: Student Rahul joins the university.
  - Update: Student changes phone number.
  - Delete: Remove graduated student's temporary hostel record.
- *Characteristics*
  - ⁕ Operates on existing data.
  - ⁕ Frequently used by applications.

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Data Query Language (DQL)

- Definition
- DQL is used to retrieve information from a database.
- *Main Command*
  - SELECT
- *Example Queries*
  - Retrieve
  - ✱ All students
  - ✱ Students of Computer Science
  - ✱ Students scoring above 80%
- *Characteristics*
  - ✱ Most frequently used language.
  - ✱ Does not modify data.
  - ✱ Supports filtering, sorting, and grouping.

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Data Control Language (DCL)

- <u>Definition</u>
- DCL controls access to database objects.
- It is primarily used by the Database Administrator (DBA).
- *Commands*
  - ⁕ GRANT
  - ⁕ REVOKE
- *Example*
  - Faculty may update marks.
  - Students may only view marks.
  - Only the Accounts Office can modify fee records.
- Importance
  - ⁕ Data security
  - ⁕ Controlled access
  - ⁕ Multi-user environment

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Transaction Control Language (TCL)

- <u>Definition</u>
- **TCL manages transactions and ensures data consistency.**
- *Commands*
  - ⁕ COMMIT
  - ⁕ ROLLBACK
  - ⁕ SAVEPOINT
- Example
  - Online Banking
  - Transfer ₹10,000
  - Debit account (A) → Credit account (B)
  - If the second operation fails, ROLLBACK restores the original balance.
- *Importance*
  - ⁕ Prevents data inconsistency.
  - ⁕ Ensures reliable transactions.

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Summary of Database Languages

Language	Purpose	Typical Commands
DDL	Define database structure	CREATE, ALTER, DROP
DML	Modify data	INSERT, UPDATE, DELETE
DQL	Retrieve data	SELECT
DCL	Control access	GRANT, REVOKE
TCL	Manage transactions	COMMIT, ROLLBACK, SAVEPOINT

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- *Memory Trick*
  - Define → DDL
  - Modify → DML
  - Query → DQL
  - Control → DCL
  - Transaction → TCL

Database Interfaces

- <u>Definition</u>
- A Database Interface is the medium through which users interact with a DBMS.
- Different users require different interfaces.
- Types
  - ⁕ Command Line Interface (CLI)
  - ⁕ Graphical User Interface (GUI)
  - ⁕ Forms-Based Interface
  - ⁕ Menu-Based Interface
  - ⁕ Natural Language Interface
  - ⁕ Web Interface
  - ⁕ Mobile Interface

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Common Database Interfaces

- Command Line Interface (CLI)
- *Example*
  - MySQL Command Prompt
  - PostgreSQL psql
- *Advantages*
  - ⁕ Powerful
  - ⁕ Fast
  - ⁕ Preferred by DBAs
- Graphical User Interface (GUI)
- *Examples*
  - ⁕ MySQL Workbench
  - ⁕ pgAdmin
  - ⁕ SQL Server Management Studio
- *Advantages*
  - ⁕ Easy to learn
  - ⁕ Visual database design
  - ⁕ Drag-and-drop tools

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Forms-Based and Menu-Based Interfaces

- Forms-Based Interface
  - Users enter data through forms.
  - *Examples*
  - ⁕ Student Admission Form
  - ⁕ Hospital Registration
  - ⁕ Railway Reservation
  - *Advantages*
  - ⁕ User-friendly
  - ⁕ Prevents invalid input
- Menu-Based Interface
  - Users choose predefined options.
  - *Examples*
  - ⁕ ATM
  - ⁕ Ticket Booking Kiosk
  - ⁕ Library Management System
  - *Advantages*
  - ⁕ Simple
  - ⁕ Suitable for beginners

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Web-Based Interfaces and APIs

- Modern applications rarely communicate directly with databases.

<u>Typical architecture:</u>

User

↓

Web Browser / Mobile App

↓

Application Server

↓

Database

- APIs (Application Programming Interfaces)
- Applications communicate with databases using APIs.
- *Examples*
  - ⁕ JDBC (Java Database Connectivity)
  - ⁕ ODBC (Open Database Connectivity)
  - ⁕ Python DB-API
  - ⁕ .NET Data Provider
- *Advantages*
  - ⁕ Secure
  - ⁕ Standardized
  - ⁕ Platform independent

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Embedded SQL and ORM (Modern Perspective)

- <u>Embedded SQL</u>
- SQL statements written inside programming languages.
- *Example*
  - Java
  - Python
  - C/C++
- Applications send SQL commands to the DBMS.
- <u>ORM (Object Relational Mapping)</u>
- ORM allows developers to work with database records as programming language objects.
- *Examples*
  - ⁕ Hibernate (Java)
  - ⁕ Entity Framework (.NET)
  - ⁕ Django ORM (Python)
  - ⁕ SQLAlchemy (Python)
- *Benefits*
  - ⁕ Less SQL code
  - ⁕ Faster development
  - ⁕ Better maintainability
- Note: Most enterprise applications use ORMs rather than writing raw SQL for every operation.

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Modern Database Interfaces (Overview)

- Modern databases support multiple access methods.
- *Examples*
  - ⁕ SQL Console
  - ⁕ REST APIs
  - ⁕ GraphQL APIs
  - ⁕ Cloud Dashboards
  - ⁕ MongoDB Compass (GUI)
  - ⁕ Mongo Shell
- <u>Cloud Databases</u>
- Cloud providers offer web-based interfaces for managing databases.
- *Examples*
  - ⁕ Amazon RDS Console
  - ⁕ Azure SQL Portal
  - ⁕ Google Cloud SQL Console
- Note: Although SQL remains the standard language for relational databases, modern systems expose databases through APIs and web services for application development.

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Real-World Case Study: Online Shopping System

Customer: Uses a website to browse products.

↓

Web Application: Processes requests.

↓

Database API (JDBC/ORM): Communicates with the database.

↓

DBMS: Executes SQL commands.

↓

Database: Stores products, customers, and orders.

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- Database Languages Used
  - ⁕ DDL → Design tables
  - ⁕ DML → Place orders
  - ⁕ DQL → Search products
  - ⁕ DCL → User permissions
  - ⁕ TCL → Complete payment transaction

Summary

- Today we learned
- ✓ Database Languages
- ✓ DDL
- ✓ DML
- ✓ DQL
- ✓ DCL
- ✓ TCL
- ✓ Database Interfaces
- ✓ CLI
- ✓ GUI
- ✓ Forms-Based Interface
- ✓ Menu-Based Interface
- ✓ APIs
- ✓ Embedded SQL
- ✓ ORM
- ✓ Modern Web & Cloud Interfaces

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Learning Reflections

- 1. Differentiate between DDL, DML, DQL, DCL, and TCL with examples.
- 2. Why is DCL important in a multi-user database?
- 3. Explain how TCL ensures consistency during banking transactions.
- 4. Compare CLI and GUI database interfaces.
- 5. What is the role of JDBC or ODBC in database applications?
- 6. Why do modern applications prefer ORMs over writing raw SQL everywhere?
- 7. Draw the architecture of a web application communicating with a database.

Homework

- Explain the five database languages–DDL, DML, DQL, DCL, and TCL–with their purposes, commands, and examples. Describe major database interfaces such as CLI, GUI, forms/menu-based, and web interfaces, and explain the role of APIs (JDBC/ODBC), embedded SQL, and ORMs in database applications.

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📌 **Heads Up!**

These slides are just **a guide** — not the full story!

To really learn , make sure to read your book, and explore other resources too.

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